Missing Values in a Markov Chain

Only the most recent piece of given information matters.

For example,

$$P(X_4 = 2 \mid X_2 = 5, X_0 = 0)$$
$$= P(X_4 = 2 \mid X_2 = 5)$$

$$P(X_4 = 2 \mid X_2 = 5, X_0 = 0) =$$

$$= \sum_{i \in S} P(X_4 = 2, X_3 = i \mid X_2 = 5, X_0 = 0)$$

$$= \sum_{i \in S} \sum_{j \in S} P(\underbrace{X_4 = 2}_{\text{event } A}, \underbrace{X_3 = i, X_1 = j}_{\text{event } B} \mid X_2 = 5, X_0 = 0)$$

"and"

Use $P(A \cap B) = P(A \mid B) \cdot P(B)$ all while staying in the "conditional universe where $X_2 = 5$ and $X_0 = 0$".

$$P(X_4=2 \mid X_2=5, X_0=0) =$$

$$= \sum_{i \in S} P(X_4=2, X_3=i \mid X_2=5, X_0=0)$$

$$= \sum_{i \in S} \sum_{j \in S} P(\underbrace{X_4=2}, \underbrace{X_3=i, X_1=j} \mid X_2=5, X_0=0)$$

$$= \sum_{i \in S} \sum_{j \in S} P(X_4=2 \mid X_3=i, \cancel{X_2=5}^{\text{M.P.}}, \cancel{X_1=j}^{\text{M.P.}}, \cancel{X_0=0}^{\text{M.P.}})$$

$$\cdot P(X_3=i, X_1=j \mid X_2=5, X_0=0)$$

So, $P(X_4 = 2 \mid X_2 = 5, X_0 = 0)$

$$= \sum_{i \in S} P(X_4 = 2 \mid X_3 = i) \sum_{j \in S} P(X_3 = i, X_1 = j \mid X_2 = 5, X_0 = 0)$$

$$= \sum_{i \in S} P(X_4 = 2 \mid X_3 = i) \cdot P(X_3 = i \mid X_2 = 5, X_0 = 0)$$

Let's move to the right-hand side!

$$P(X_4 = 2 \mid X_2 = 5)$$

$$= \sum_{i \in S} \sum_{j \in S} \sum_{k \in S} P(X_4 = 2, X_3 = i, X_1 = j, X_0 = k \mid X_2 = 5)$$

$$= \sum_{i \in S} \sum_{j \in S} \sum_{k \in S} P(X_4 = 2 \mid X_3 = i, \overset{\text{M.P.}}{\cancel{X_2 = 5}}, \overset{\text{M.P.}}{\cancel{X_1 = j}}, \overset{\text{M.P.}}{\cancel{X_0 = k}})$$

$$\cdot P(X_3 = i, X_1 = j, X_0 = k \mid X_2 = 5)$$

$$= \sum_{i \in S} P(X_4 = 2 \mid X_3 = i) \cdot \underbrace{P(X_3 = i \mid X_2 = 5)}$$

Need this to equal $P(X_3 = i \mid X_2 = 5, X_0 = 0)$

$$P(X_3 = i \mid X_2 = 5, X_0 = 0)$$

$$= \sum_{j \in S} P(X_3 = i, X_1 = j \mid X_2 = 5, X_0 = 0)$$

$$= \sum_{j \in S} P(X_3 = i \mid X_2 = 5, \cancel{X_1 = j}^{\text{M.P.}}, \cancel{X_0 = 0}^{\text{M.P.}}) \cdot P(X_1 = j \mid X_2 = 5, X_0 = 0)$$

$$= P(X_3 = i \mid X_2 = 5) \cdot 1 = P(X_3 = i \mid X_2 = 5) \checkmark$$

Conditional Independence

Let $\{X_n\}$ be a Markov chain.

We have, for example, that

$$P(X_4 = 1 \mid X_3 = 1, X_2 = 2) = P(X_4 = 1 \mid X_3 = 1)$$

<u>Question</u>: Does this mean that X_4 is independent of X_2?

<u>Answer</u>: No. They are <u>conditionally independent</u> given X_3.

We can specifically use the Markov property to show that

$$P(X_4 = x_4, X_2 = x_2 \mid X_3 = x_3)$$

$$= P(X_4 = x_4 \mid X_3 = x_3) \cdot P(X_4 = x_4 \mid X_3 = x_3)$$